

Total Number of Pages: 01

Student Name : _____

Enrollment Number: _____

Mid Term Examination- Nov 2022

Program : B. Tech

Paper Code : ECC 205

Time Duration : $1\frac{1}{2}$ hours

Semester : Third

Paper Name : Signals and Systems

Maximum Marks : 30

Note: Question No. 1 is compulsory. Attempt any two questions from the remaining questions. Only scientific calculator is allowed.

Question 1

- 1(a) Check whether the system described by $y(n) = x(n) + 6$ is linear or non-linear. [2]
- 1(b) Sketch $x(-2t + 1)$ for a given signal $x(t) = 2[u(t) - u(t - 3)]$. [2]
- 1(c) Determine whether a signal $x(t) = e^{-\alpha|t|}$, $\alpha > 0$ is energy or power signal or neither. [2]
- 1(d) Determine the Nyquist rate of a signal $x(t) = \frac{\sin(4000\pi t)}{\pi t}$. [2]
- 1(e) Obtain the Fourier series coefficients of a discrete time signal $x(n) = 0.5 + \cos(3\pi n + \pi/3)$. [2]

Question 2

- 2(a) Find the output of an LTI systems whose impulse response is $h(t) = e^{-2t}u(t)$ and input is $x(t) = e^{-t+1}u(t-1)$. [5]
- 2(b) Find the linear convolution of two sequences $x_1(n) = [2, 1, -1, 3]$ and $x_2(n) = [2, 1, 5]$. [5]

Question 3

- 3(a) Determine a signal $x(t)$, when the following facts about $x(t)$ are given:
1. $x(t)$ is a real and odd signal.
 2. $x(t)$ is periodic with period $T = 4$, and it has exponential Fourier series coefficients c_k .
 3. $c_k = 0$ for $|k| > 1$.
 4. $\frac{1}{4} \int_0^4 |x(t)|^2 dt = 4$.
- [5]
- 3(b) Consider a causal LTI system with frequency response $H(\omega) = \frac{1}{3+j\omega}$. For a particular input $x(t)$, this system is observed to produce the output $y(t) = e^{-3t}u(t) - e^{-4t}u(t)$. Determine the input $x(t)$. [3]

Question 4

- 4(a) Find the Fourier transform of a signal $x(t) = \frac{1}{1+e^t}$. However, the Fourier transform of $g(t) = e^{-|t|}$ is given as: $G(\omega) = \frac{2}{1+\omega^2}$. [5]
- 4(b) The Laplace transform of a signal $x(t)$ is given as $X(s) = \frac{3s+1}{s^2+4s^2+(k-3)s}$. If $\lim_{t \rightarrow \infty} x(t) = 1$, then find the value of k ? [5]

END TERM EXAMINATION

THIRD SEMESTER [B.TECH.] FEBRUARY 2023

Paper Code: ECC205**Subject: Signals and Systems****Time: 3 Hours****Maximum Marks: 75**

**Note: Attempt five questions in all including Q.No.1 which is compulsory.
Select one question from each unit.**

Q1 Attempt **all** questions: (3x5=15)

- Find the period of the signal $x(n) = e^{(-1+j)t}$
- What is Nyquist Criteria? Write down the equation.
- Explain Gibbs Phenomenon in brief.
- What is the difference between DTFT and DFT?
- Write properties of ROC for Z-Transform.

UNIT-I

Q2 a) Compute whether the following systems is (5)
 $y(n) = x(n) + 2$

- Causal or non-causal
- Time variant or time invariant

b) Draw waveform for the following signals: (10)

- $y(t) = u(-t + 2)$
- $y(t) = r(3t - 2)$
- $y(t) = r(t) - r(t - 1)$

OR

Q3 What are the Dirichlet conditions for Fouries Series? Write down and prove any 5 properties of Fourier Series. (15)

UNIT-II

Q4 Find the Fourier Transform of (15)

- $x(t) = \cos(wt)u(t)$,
- $x(t) = t \exp(-at)u(t)$

OR

Q5 Find the final value of the response $y(t)$ of a system whose transfer function is

$$H(s) = \frac{s+3}{s^2+4s+5}$$

When the system is excited by

- Unit step
 - Unit impulse
- (15)

P.T.O.

[-2-]

UNIT-III

- Q6 a) Compute the DFT of the following sequence (10+5)
 $x(n) = \{1, 2, 0, 1\}$
b) What is twiddle factor? Write down the properties.

OR

- Q7 a) Compute circular convolution of the following signals: (7.5x2)
 $x(n) = \{1, 2, 1, 2\}$
 $h(n) = \{1, 0, 1, 0\}$
b) Compute DFT of the following using DIT FFT
 $x(n) = \{1, 2, 0, 1, 0, 0, 0, 1\}$

UNIT-IV

- Q8 a) Determine the Z-transform and ROC of the following signal (7.5x2)
 $x(n) = a^n u(n) - a^n u(n-1)$
b) Find inverse z-transform of the following signal
 $H(Z) = \frac{1}{1-2z^{-1}} \quad |z| > 2$

OR

- Q9 Consider the system described by following equation: (15)
 $y(n) - \frac{1}{2}y(n-1) = x(n) + 2x(n-1)$
Compute a. Impulse response b. Step Response

MID TERM EXAMINATION

B.TECH ECE, EEE

3rd Semester, November 2023

Paper Code: ECC-205

Subject: Signals and Systems

Time: 1½ Hrs.

Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q.1.

a) Sketch the following signals:

(2) (CO 1)

I. $2u(-n+2)$

II. $r(-t+2)$

b) Find the odd and even components of the signal $x(t) = e^{-2t}u(t)$

(2) (CO 1)

c) Define Aliasing and how is it removed.

(2) (CO 1)

d) Find the Fourier Transform of $x(t) = e^{-3|t|}$

(2)(CO 2)

e) Compute whether the following discrete time system is linear, time invariant and stable
 $y(n) = n x(n)$

(2) (CO 2)

Q.2. (a) Determine the Fourier series coefficients for the following periodic discrete time signal.

(5) (CO 1)

$$x(n) = \cos \frac{2\pi n}{3} + \sin \frac{2\pi n}{7}$$

Plot the magnitude and phase of each set of coefficients c_k .

(b) Consider an analog signal $x(t) = 5 \cos 200\pi t$.

Examine the minimum sampling rate to avoid aliasing. If the sampling rate is $F_s=400\text{Hz}$, what will be the discrete time signal after sampling?

(5) (CO 1)

Q.3. (a). The output of a causal LTI system is related to the input $x(t)$ by the differential equation

$$\frac{dy(t)}{dt} + 2y(t) = x(t)$$

Find $y(t)$ for the input $x(t) = e^{-t}u(t)$

(5) (CO 2)

(b) Find the output of the discrete time LTI system for the input $x(n) = \left(\frac{1}{2}\right)^n u(n)$ and $h(n) = u(n)$

(5) (CO 2)

Q.4. (a) If a LTI system is defined by transfer function $H(s) = \frac{s-1}{s^2+s-12}$. Find the impulse response for the following conditions

(5) (CO 2)

I. If the system is stable

II. If the system is causal

(b) Check whether the following signal is periodic, Energy /Power signal.

$$x(n) = e^{j\frac{2\pi}{5}(n+\frac{1}{2})}$$

(5) (CO 1)

If it is periodic, find the period of the signal and also find out the energy/power of the signal.

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY 2024

Paper Code: ECC-205

Subject: Signals and Systems

Time: 3 Hours

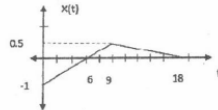
Maximum Marks: 75

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Assume missing data if any.

- Q1 Attempt all questions:-
 (a) Classify the Following signals a1) Even and Odd a2) Periodic and Aperiodic a3) Energy and Power signals (3)
 (b) Define the properties of ROC for Laplace Transform. (3)
 (c) Calculate number of additions and multiplications for N point DFT and FFT for N=16. (3)
 (d) Define the Causality and Stability criteria using Z Transform. (3)
 (e) Define Parseval's Relation. (3)

UNIT-I

- Q2 (a) Describe impulse, step and ramp signals. (3)
 (b) Check whether energy or power signal:
 B1) $x(t) = e^{j(2t + \frac{\pi}{2})}$ B2) $x(n) = \cos((1/4)n)$ (3)
 (c) Find the fundamental period of following signals.
 C1) $x(t) = \cos(10\pi t + 1) - \sin(4\pi t - 1)$ C2) $x(n) = \sin(3\pi n + 0.6\pi n)$ (3)
 (d) Perform the following transformations on given signal. (6)



- a) $x(-t)$ b) $x(t+6)$ c) $x(3t)$ d) $x(3t+6)$

- Q3 (a) Find the Fourier Series Coefficients of the Following Function: (5)
 $x(t) = 1 + \sin(\cos t) + 2 \cos(\cos t) + \cos(2\cos t + \pi/4)$
 (b) Find the Fourier series Coefficients for: (10)
 1) $x(n) = \cos \sqrt{2} \pi n$ 2) $x(n) = \cos \pi n/3$ 3) $x(n) = [1 \ 1 \ 0 \ 0]$

UNIT-II

- Q4 (a) Find the Fourier Transform of a signal $x(t) = \frac{2}{1+t^2}$. However, the Fourier Transform of $g(t) = e^{-a|t|}$ is given as $G(\omega) = \frac{2a}{1+\omega^2}$ (7.5)
 (b) An LTI system is described by the differential equation (7.5)
 $\frac{dy(t)}{dx} + (t) = (t)$, Find the impulse response of the system

P.T.O.

[-2-]

- Q5 (a) Determine whether system $y(n) = x(n+5) - x(n-7)$ is linear or not. (5)
 (b) Determine whether system $y(t) = t^2 x(t-4)$ is time invariant. (5)
 (c) Find the Laplace Transform of the following Signals: (5)
 1) Unit Step Signal $u(t)$ 2) Delayed Unit Step $Ku(t-a)$
 3) Ramp Signal $r(t)$

UNIT-III

- Q6 An 8-point sequence is given by $x(n) = [2, 2, 2, 2, 1, 1, 1, 1]$. Compute 8 point DFT of $x(n)$ by radix 2 DIF FFT. (15)
 Q7 (a) Perform the circular convolution of following sequences: (7.5)
 $x(n) = [1 \ 2 \ 2 \ 3]$ and $h(n) = [1 \ 3 \ 2 \ 1]$
 (b) Perform the N point DFT on the following sequence: (7.5)
 $x(n) = [1, 2, 3, 3]$ and also plot magnitude & Phase plots.

UNIT-IV

- Q8 (a) Define any Four properties of Z Transforms. (7.5)
 (b) Find the Z transform of Following Signals: (7.5)
 1) $x(n) = 2^n U[n]$ 2) $x(n) = -2^n U[-n-1]$
 Q9 Find the Inverse Z Transform of the function $X(Z) = \frac{1}{1-1.5Z^{-1}+0.5Z^{-2}}$ by using 1) Long Division method 2) Partial Fraction Expansion, (15)
 a) when ROC: $|Z| > 1$ b) when ROC $|Z| < 1/2$

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] DECEMBER 2024

Paper Code: ECC-205

Subject: Signals and Systems

Time: 3 Hours

Maximum Marks: 60

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

- Q1 Attempt all questions:
 (a) Check whether the signal $x(t) = 2 \cos(50\pi t) + \sin(10\pi t)$ is periodic or aperiodic. If periodic, determine the fundamental period. (4)
 (b) A discrete time signal is described by $x[n] = [1, 2, 3, 2, 1]$. Determine $x[n/2]$. (4)
 (c) Find Discrete Time Fourier Transform of signal $x[n] = 2^n u[n]$. (4)
 (d) State and prove final value theorem of Z-Transform. (4)
 (e) Write the Dirichlet conditions for existence of Fourier Transform. (4)

UNIT-I

- Q2 (a) A continuous time signal is expressed by $x(t) = 1 + \sin \omega_0 t + 2 \cos \omega_0 t + \cos(2\omega_0 t + \frac{\pi}{4})$, where ω_0 is the fundamental frequency of the signal. Determine the Fourier series coefficients of $x(t)$. Also, sketch their magnitude plot. (7)
 (b) State and explain sampling theorem (3)

OR

- Q3 Let $x(t) = u(t-3) - u(t-5)$ and $h(t) = e^{-3t} u(t)$. Compute (10)
 a) $y(t) = x(t) * h(t)$
 b) $g(t) = [dx(t)/dt] * h(t)$
 c) How is $g(t)$ related to $y(t)$?

UNIT-II

- Q4 (a) Consider a causal LTI system with frequency response $H(\omega) = \frac{1}{1+j\omega}$. For a particular input $x(t)$ this system is observed to produce the output $y(t) = e^{-3t} u(t) - e^{-4t} u(t)$. Determine $x(t)$. (7)
 (b) Find the Fourier Transform of signal $x(t) = \frac{1}{1+t^2}$. However, the Fourier Transform of $g(t) = e^{-|t|}$ is given as: $G(\omega) = \frac{2}{1+\omega^2}$. (3)

OR

- Q5 (a) Determine whether the signal $y(t) = x(t) * x(t-2)$ is (7)
 (a) Linear or nonlinear, (b) Static or dynamic, (c) Causal or non-causal and (d) Time variant or time invariant.

P.T.O.

[-2-]

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- (b) Consider an LTI system with the differential equation (3)
 $\frac{d^2 y(t)}{dt^2} + 4 \frac{dy(t)}{dt} + 3y(t) = \frac{dx(t)}{dt} + 2x(t)$. Find the frequency response & impulse response.

UNIT-III

- Q6 (a) Compute the Discrete Time Fourier Transform of the following functions. (7)
 i. $x[n] = u[n-2] - u[n-6]$
 ii. $x[n] = (1/2)^n u[n-1]$

- (b) Find Inverse Discrete Time Fourier Transform of a signal (3)
 $X(e^{j\omega}) = \frac{e^{-j\omega} - \frac{1}{5}}{1 - \frac{1}{5}e^{-j\omega}}$

OR

- Q7 (a) Find the circular convolution of the following sequences: (7)
 $x_1[n] = [1, -1, 2, 3]$; $x_2[n] = [0, 1, 2, 3]$
 (b) Compute 4-point DFT of $x[n] = [1, 2, 1, 2]$ (3)

UNIT-IV

- Q8 (a) Determine the Z-Transform and ROC of the signal (7)
 $x[n] = 7(\frac{1}{2})^n u[n] - 6(\frac{1}{2})^n u[n]$
 (b) Write the properties of Region of Convergence. (3)

OR

- Q9 Consider a signal $y[n]$ which is related to two signals $x_1[n]$ and $x_2[n]$ (10)
 by

$$y[n] = x_1[n+3] * x_2[-n+1]$$

$$\text{Where } x_1[n] = (\frac{1}{2})^n u[n] \text{ and } x_2[n] = (\frac{1}{3})^n u[n]$$

Given that

$$a^n u[n] \xrightarrow{Z} \frac{1}{1-az^{-1}}, |z| > |a|$$

Use the properties of Z-Transform to determine the Z-Transform $Y(z)$ of $y[n]$.

P-1/2
ECC-205P-2/2
ECC-205